



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Sequence in Metric Space

Convergence of Sequences, Cauchy Sequences, Subsequences

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CONDUCTED BY

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Sequence in Metric Space

Sequence in Metric Space

Let (X, d) be a metric space. A sequence in X is a function $f : \mathbb{N} \rightarrow X$. We usually denote the sequence by $\{x_n\}$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

Thank You!

We'd love your questions and feedback.

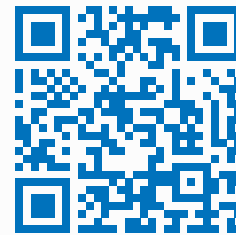
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.